Freight Rate Prediction — Assessment Report

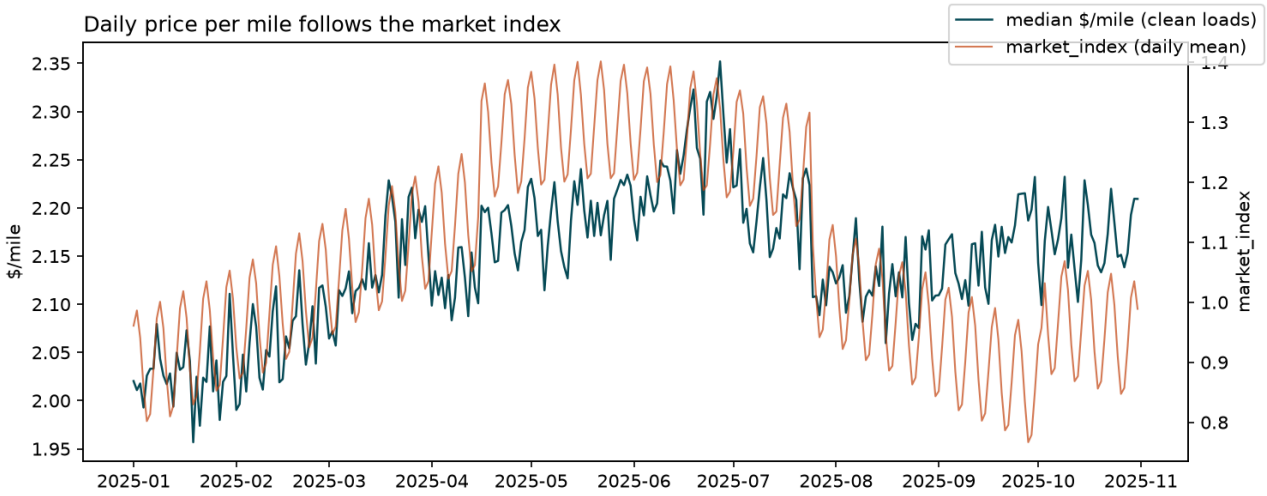
Machine Learning Engineer assessment · Spotter · Gurjit Singh

1. Summary

A LightGBM model predicts each load's rate per mile, multiplied back by distance. On a strict time-based split it reaches 1.97% MAPE (MAE $46) on the latest month (Oct 2025) and 2.38% MAPE on loads involving cities it never saw in training. Removing ~1.3% corrupt labels was the single biggest improvement (holdout MAPE 4.2% → 2.3%).

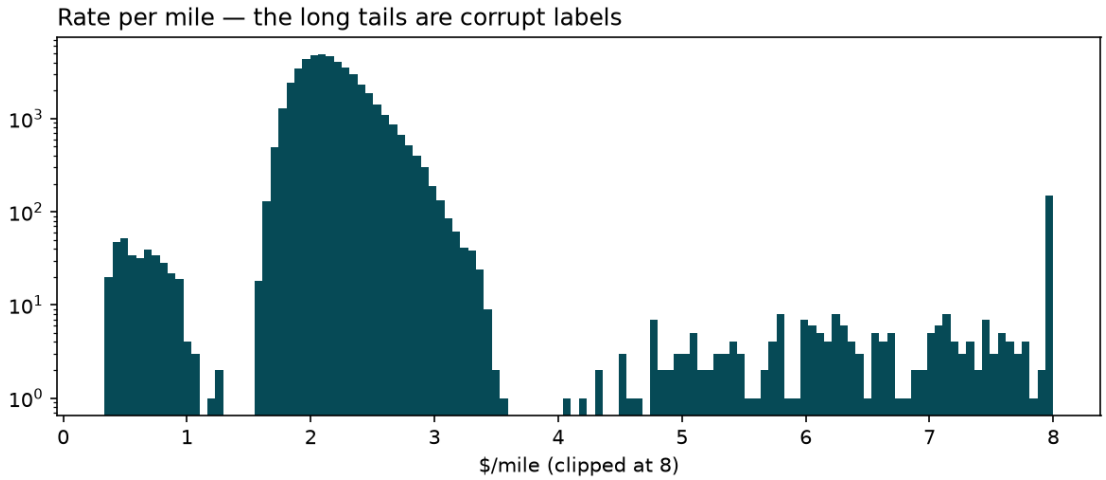
2. Data and key findings

Training data: 48,000 labelled loads from 2025-01-01 to 2025-10-31. Validation: 12,000 loads from 2025-11-01 to 2025-12-31 — entirely in the future relative to training. Distance explains most of the price (correlation 0.91 with rate). Reefer is the most expensive equipment per mile, then Flatbed, then Dry Van. market_index is essentially one value per day and tracks the daily median price per mile (correlation ≈ 0.7), with a weekly cycle. Eight cities in the validation set never appear in training, so the model must generalise beyond city names.



3. Data-quality issues and how they were handled

Issue	Evidence	Handling
Corrupt rate labels	671 rows: $/mile 2–7x off	Removed from training only
Negative weights	292 rows (down to -47,500 lb)	Absolute value (sign error)
Missing weight	300 rows	Median weight for that equipment
Missing market_index	374 rows	That day's mean (index is per-day)
Coordinates vs distance	Short lanes: coords drift	Trust given road distance
Unseen cities	8 cities only in validation	Coordinate features, no city IDs



One cleaning idea was tested and rejected: rebuilding distances from coordinates where they disagreed. Those rows were all short lanes (~70–90 miles) whose road distance is consistent across loads on the same lane — the synthetic coordinates were the noisy part — so the 'fix' made predictions worse and was removed.

4. Train / validation split

Because the final validation set (Nov–Dec 2025) lies after all training data (Jan–Oct 2025), a random split would leak future market conditions and overstate accuracy. I used an expanding-window time split: for each test month M (Jul, Aug, Sep, Oct), the model is trained on every earlier month and evaluated on M. Errors are reported on clean rows (outlier labels excluded, since no model can predict a data-entry error) and on all rows. A second test removes 8 random cities from training and evaluates Sep–Oct loads that touch them, mirroring the unseen cities in the validation file. Label-outlier rules are fitted on the training part of each split only.

Test month	MAPE clean	MAE clean	MAE all rows	Rows
2025-07	1.19%	$28	$83	4912
2025-08	5.71%	$112	$160	4759
2025-09	2.1%	$51	$102	4670
2025-10	1.97%	$46	$105	4853
Unseen cities	2.38%	$59	$116	2516

August is the weak month (5.7% MAPE): the market index fell sharply from July and prices adjusted differently than earlier in the year, a regime change that models trained only on Jan–Jul could not anticipate. Adding 7- and 28-day rolling market features was tested; it did not fix August and degraded later months (Oct 1.97% → 4.67%), so it was not used.

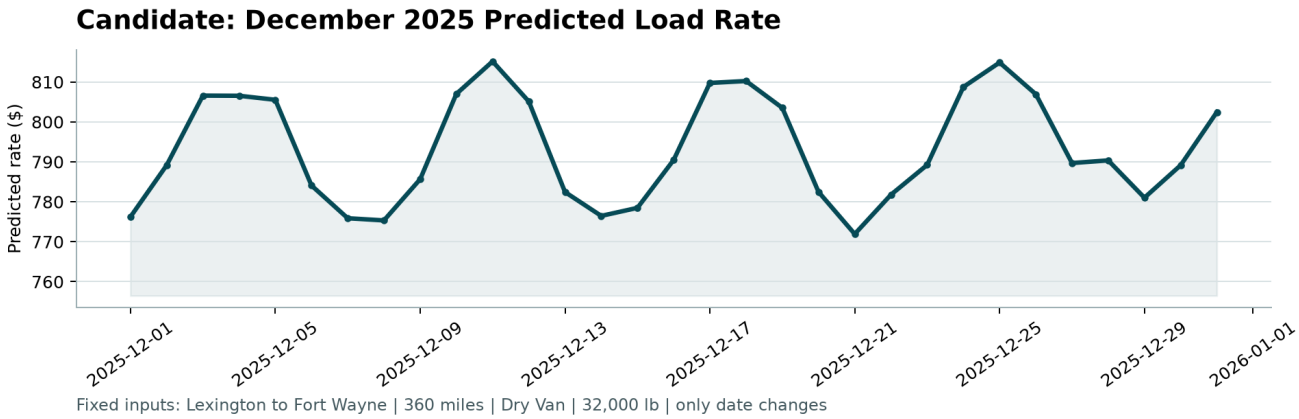
5. Model choice

Model (train Jan–Aug, test Sep–Oct)	MAPE clean	MAE clean
Median $/mile by equipment x distance	8.15%	$176
Ridge regression on log(rate)	4.54%	$106
LightGBM on log(rate)	2.60%	$59
LightGBM on rate per mile (chosen)	2.31%	$54
Same, but corrupt labels kept	4.23%	$97

Gradient-boosted trees capture the non-linear interplay of distance, equipment, region and market level without manual interactions. Predicting rate per mile makes the target comparable across short and long hauls, so the trees spend their capacity on the price level rather than on re-learning distance. Top features by gain: straight_miles (30%), log_distance (22%), eq_dry_van (14%), quote_signal (13%), distance (9%), eq_reefer (3%).

6. December 2025 prediction chart (from score.py)

Fixed load: Lexington → Fort Wayne, 360 miles, Dry Van, 32,000 lb; only the date changes. The December inputs have no coordinates or market signals, so city coordinates come from the training data and each day's market_index / quote_signal mean comes from the validation file, which already covers December. Predictions range $772–$815 (≈ $2.15–2.26 per mile) with the weekly market rhythm; historical Dry Van loads on this lane averaged $808.



7. Limitations and next steps

• Regime shifts (like August) remain the main error source — external market data (diesel prices, load-to-truck ratios) would help.

• Lane-level history could be added as a smoothed target encoding with time-aware cross-fitting.

• Prediction intervals (quantile LightGBM) would let pricing teams see uncertainty, not just a point estimate.